

Deep Subsurface Core Drilling Telemetry & Borehole Stratigraphy

1. Rotary Percussive Drill Mechanics

The 10-meter coring rig operates at 400 RPM with 15 Hz axial percussion, utilizing polycrystalline diamond compact (PDC) drill bits cooled by pressurized gaseous nitrogen.

2. Subsurface Ice Layer Detection

At borehole depth 4.8 meters, drill telemetry registered torque drop from 45 Nm to 12 Nm, confirming penetration into massive glacial ice sheet with 97 percent water purity.

3. Core Extraction Logging

Extracted 25 mm diameter stratigraphy cores are cataloged into automated cryogenic sample cassettes preserved at 210 K to avoid thermal alteration.